Polynomially many total mistakes for Bayesian neural-network learning

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Abstract

For uniformly sampled Boolean inputs, the strong-prior, above-mean-field, and mean-field learning results imply a stronger conclusion than eventual fixed accuracy: the total number of mistakes over an infinite learning stream has a polynomial bound, independent of learner width. The bound holds in expectation and with high probability, and applies both to posterior sampling and to counting a prediction as a failure unless the correct label has posterior probability greater than 2/3. The new statistical argument works for any consistency posterior whose relevant functions have a short finite description, even if their prior weights are extremely unequal. Deleting repeated inputs reduces the infinite iid stream to a uniformly random permutation. A harmonic bound over its random prefixes then gives an overhead of order the input dimension. We state precisely the structural facts imported from the neural-network papers, give explicit budgets, and distinguish this result from an arbitrary-order mistake guarantee. A prediction-preserving neuron-merging restriction also retains polynomial dependence on depth. These new cumulative conclusions have not been Lean verified.

1 Prediction protocol and the result

Let $\mathcal{X}$ be a finite set of size $m \geq 2$, let $f : \mathcal{X} \to \{-1, 1\}$, and let π be a prior on Boolean functions with $\pi(f) > 0$. Before round j, the learner has seen the true labels of the preceding inputs and forms the exact consistency posterior Π_{j-1}. The next input X_j is then presented, a prediction is made, and its true label $f(X_j)$ is revealed. Every label is revealed, including after a correct prediction. Define

$$q_j = \Pi_{j-1}\{g : g(X_j) \neq f(X_j)\}, \qquad S = \sum_j q_j.$$

Thus S is cumulative posterior error probability. If the learner draws a function from the current posterior at each round, its total number of incorrect labels is denoted Z. The expected values of Z and S agree. For a fixed $0 < \theta < 1$, define the confidence-failure count

$$F_\theta = \sum_j \mathbf{1}\{\Pi_{j-1}(g(X_j) = f(X_j)) \leq \theta\}.$$

The $> 2/3$ confidence convention is $F_{2/3}$. Pointwise,

$$F_\theta \leq S/(1 - \theta). \tag{1}$$

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Any majority classifier's number of errors is at most $2S$, including arbitrary decisions at ties. These inequalities allow posterior mixtures; they require neither a unique minimum-complexity function nor posterior concentration on one function.

We consider either a uniformly random permutation of $\mathcal{X}$, used once, or an infinite iid stream uniform on $\mathcal{X}$. In the latter protocol every repeated input has posterior correctness probability one, and duplicating a hard conditioning constraint leaves the posterior unchanged. The order of first appearances is a uniformly random permutation. Consequently the total variables S, F_θ, Z have the same laws in the two protocols (using fresh posterior randomness on first appearances). We can therefore prove all bounds for m rounds without replacement.

For $\mathcal{X} = \{-1, 1\}^d$, the neural-network consequences are:

- At every fixed depth, the strong-prior learner has a total mistake bound polynomial in d and bounded-weight target-network complexity, at every width above that complexity, once its variance is sufficiently small at that fixed width.
- At every fixed depth and every fixed $\gamma > 1$, the specified finite-variance learner has the same type of polynomial bound, uniformly over all admissible widths. Its polynomial may depend on γ.
- The finite-variance mean-field tanh learner, $\gamma = 1$, also has the same polynomial bound at every admissible width.

High-probability bounds are polynomial in $\log(1/\delta)$ as well. The theorem does not replace uniform random input order by arbitrary order.

2 A cumulative bound from a finite relevant class

Write $\mathsf{H}_r = \sum_{k=1}^{r} 1/k$. The following theorem is the new statistical ingredient. In particular, it does not bound mistakes by the negative log prior probability of the target.

Theorem 1 (Random-order cumulative errors)**.** *Suppose A is a finite set of Boolean functions containing f and*

$$\log |A| \le h, \qquad \pi(A^c) \le \beta\pi(f), \qquad 0 \le \beta < 1. \tag{2}$$

Under either uniform input protocol above,

$$\mathbb{E}S = \mathbb{E}Z \le B_E := 1 + 2(h+1)\mathsf{H}_{m-1} + m\beta. \tag{3}$$

For $0 < \delta < 1$, put

$$a = h + \log(2m/\delta), \qquad B_\delta = 1 + 2a\mathsf{H}_{m-1} + m\beta, \qquad U_\delta = (e-1)B_\delta + \log(2/\delta).$$

Then $\mathbb{P}\{S > U_\delta\} \le \delta$ and $\mathbb{P}\{Z > U_\delta\} \le \delta$. These are separate probability statements; replace δ by $\delta/2$ to require both simultaneously. Equation (1) gives the corresponding bounds for every confidence threshold and for majority errors.

Proof. After any correctly labeled history, the posterior denominator is at least $\pi(f)$. Hence the posterior mass outside A is at most β, uniformly over all histories.

Let S_n now denote the set of the first n inputs of the permutation, and let

$$R_n = \max_{g \in A:\, g = f \text{ on } S_n} \frac{1}{m} |\{x : g(x) \ne f(x)\}|.$$

The maximum exists because f is included. A function with k errors survives a uniform size-n prefix with probability

$$\frac{\binom{m-k}{n}}{\binom{m}{n}} \le (1-k/m)^n \le e^{-nk/m},$$

with survival probability zero when $n > m-k$. A union bound gives

$$\mathbb{P}\{R_n > u\} \le \min\{1, e^{h-nu}\}, \qquad \mathbb{E}R_n \le \frac{h+1}{n} \quad (1 \le n < m). \tag{4}$$

All errors of a surviving function are among the remaining $m-n$ inputs. Conditional on the history, the expected posterior error on the next uniformly chosen remaining input is therefore at most

$$\frac{m}{m-n} R_n + \beta. \tag{5}$$

The first round costs at most one. Summing expectations and using

$$\sum_{n=1}^{m-1} \frac{m}{n(m-n)} = 2\mathsf{H}_{m-1}$$

proves (3). For posterior sampling, condition additionally on the current input before taking the expectation of the sampled label.

For high probability, let $E_n = \{R_n \le a/n\}$. By (4), the probability that any E_n fails is at most $(m-1)e^{h-a} \le \delta/2$. Stop counting at the first failed prefix event. More precisely, replace the round-$n+1$ variable Y_{n+1} by $Y'_{n+1} = Y_{n+1}\mathbf{1}\{E_1, \ldots, E_n\}$, with the indicator one at $n = 0$, where Y is either q or the sampled error indicator. The indicator is known before the next input. The sum of conditional expectations of Y' is at most B_δ, by (5). For any adapted variables $0 \le Y'_j \le 1$, the elementary bound

$$\mathbb{E}[e^{Y'_j} \mid \mathcal{F}_{j-1}] \le 1 + (e-1)\mathbb{E}[Y'_j \mid \mathcal{F}_{j-1}] \le \exp((e-1)\mathbb{E}[Y'_j \mid \mathcal{F}_{j-1}])$$

shows that $\exp(\sum_j Y'_j - (e-1)\sum_j \mathbb{E}[Y'_j \mid \mathcal{F}_{j-1}])$ is a nonnegative supermartingale. Markov's inequality bounds $\mathbb{P}\{\sum Y'_j > U_\delta\}$ by $\delta/2$. On all good prefixes, $\sum Y'_j = \sum Y_j$; a union bound proves the claim. □

For $m = 2^d$, take $\beta = 1/m$. Then $\mathsf{H}_{m-1} \le 1 + d\log 2$, so

$$B_E \le 2 + 2(h+1)(1 + d\log 2), \qquad U_\delta = O((h + d + \log(1/\delta))(d+1)). \tag{6}$$

The relevant description bound need only remain polynomial when $\log(1/\beta) = d\log 2$; it need not be independent of the desired tail.

What is uniform in width? The numerical bounds below hold for each admissible learner width with the same constants. The high-probability theorem is applied to a fixed learner and its posterior randomness. It does not assert that one event simultaneously controls independently sampled predictions of every learner.

3 The direct prior-mass bound

The smaller-width branches of the finite-variance neural-network proofs use a different, simpler estimate.

Lemma 2 (Exact Bayesian telescoping)**.** *For every input sequence, including adaptively chosen inputs,*

$$S \le \sum_j -\log(1-q_j) \le -\log \pi(f) =: J.$$

Thus $F_\theta \le J/(1-\theta)$ deterministically. Posterior sampling satisfies $\mathbb{E}Z \le J$, and

$$\mathbb{P}\{Z > (e-1)J + \log(1/\delta)\} \le \delta.$$

Proof. The product $\prod_j (1-q_j)$ is the prior probability that a function fits all revealed labels, and is at least $\pi(f)$. Use $q \le -\log(1-q)$. The sampled-error assertion follows from the same exponential supermartingale as above, conditional on each revealed query. Finite-horizon bounds extend to infinite streams by monotone convergence. □

This lemma alone generally costs a factor growing with neural-network width and inverse prior variance. The relevant-class theorem avoids that factor at large widths; the lemma covers the remaining widths where its cost is already polynomial.

4 Application to the three neural-network regimes

4.1 Model and imported structural results

The domain is $\{-1,1\}^d$. A depth-L network has $L-1$ hidden layers of width N, a constant input coordinate, bounded 1-Lipschitz hidden activation, and a linear readout. Define

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1}\|W^s\|_F^2 + N\|W^L\|_2^2.$$

The raw Gaussian is conditioned, before training, on $|z_w(x)| > 1/2$ for every input. It therefore induces a prior on Boolean sign functions. Training uses exact label agreement. Let $C = C_L(f)$ be the least width of a depth-L implementation with strict margin $1/2$ and all weights bounded in magnitude by that width.

We use the following proved structural facts from [1, 2, 3]. These are imports, not newly established Gaussian-compression theorems. Their explicit scalar budgets appear in Appendix A.

1. There is a common finite set containing every sign function whose minimum weighted energy is at most J, with log cardinality at most $F_L(d,J)$. This is polynomial in d, J at fixed L. Cloning gives $\kappa_N(f) \le B_L(d,C) := d(d+1)C^2 + LC^4$.

2. A raw target neighborhood has mass at least $\exp(-AN^\gamma/2)$ for every $N \ge C$, where $A = A_L(d,C) \ge 1$ is polynomial at fixed depth and the Gaussian variances are $N^{1-\gamma}/d, N^{-\gamma}, N^{-\gamma-1}$.

3. At fixed $\gamma > 1$, for $K \ge 1$ and widths satisfying

$$N^{\gamma-1} \ge 8(d+1), \qquad N^{\gamma-1} \ge 64 a_K^{-2}, \qquad a_K = [8L(\sqrt{K}+1)^{L-2}]^{-1},$$

the raw total mass of full-margin functions with minimum energy greater than $(L+3)K$ is at most $Le^{-KN^\gamma/8}$.

4. For tanh at $\gamma = 1$, if $K \geq 4$ and $N \geq T_L(d, K)$, the raw total mass of full-margin functions outside a common finite class of log cardinality at most $G_L(d, K)$ is at most $Le^{-NK/8}$. Both T_L and G_L are polynomial at fixed depth.

The last two bounds concern total exceptional mass, not one exceptional function at a time. This distinction is essential for this application.

4.2 Strong-prior limit

In the strong-prior parameterization, the energy coordinates are independent centered Gaussians of variance σ^2. At fixed width,

$$2\sigma^2 \log \mu_{N,\sigma}\{w : g(x)z_w(x) > 1/2 \ \forall x\} \longrightarrow -\kappa_N(g).$$

There are finitely many Boolean functions. Hence, with $A_{N,f} = \{g : \kappa_N(g) \leq \kappa_N(f)\}$,

$$\frac{\pi_{N,\sigma}(A^c_{N,f})}{\pi_{N,\sigma}(f)} \longrightarrow 0.$$

All functions tied with the target's energy remain in this set. By cloning and compression, its log cardinality is at most $h = F_L(d, B_L(d, C))$.

Corollary 3 (Strong-prior cumulative learning)**.** *At every fixed L, for every $N \geq C$ there is $\sigma_0 > 0$ such that all $0 < \sigma < \sigma_0$ satisfy Theorem 1 with $m = 2^d$, $\beta = 2^{-d}$, and $h = F_L(d, B_L(d, C))$. Consequently the total expected and high-probability mistake bounds are polynomial in d, C (and $\log(1/\delta)$ for high probability), independently of N. The threshold σ_0 may depend on N, d, L, f and the activation; it need not be uniform in width. One fixed sufficiently small variance works for the entire infinite stream.*

The same bounds hold for any limit of these finite-domain posterior laws as $\sigma \downarrow 0$. No selection of a unique energy minimizer is required. Ordinary function-MAP prediction also obeys the corresponding finite-class random-order bound: when $\beta < 1$, every function outside $A_{N,f}$ has smaller prior mass than the consistent target and cannot be MAP. Ties must be selected independently of the next input. Parameter MAP and a choice of an unattained minimum-energy parameter are different notions and are not identified with function MAP here.

4.3 Every fixed exponent $\gamma > 1$

Set $m = 2^d$, $A = A_L(d, C)$, and

$$\begin{aligned} K &= 4A + 8\log(Lm), \qquad a = a_K, \\ T &= \left\lceil \max\{C, [8(d+1)]^{1/(\gamma-1)}, (8/a)^{2/(\gamma-1)}\} \right\rceil, \\ h &= F_L(d, (L+3)K), \qquad J = AT^\gamma/2. \end{aligned}$$

At $N \geq T$, the relative exceptional mass is at most

$$L\exp\{-(K/8 - A/2)N^\gamma\} = L(Lm)^{-N^\gamma} \leq 1/m.$$

The corresponding finite set necessarily contains f, since otherwise the ratio would be at least one. At $C \leq N < T$, $-\log \pi(f) \leq AN^\gamma/2 \leq J$.

Corollary 4 (Above-mean-field cumulative learning)**.** *For every fixed L and $\gamma > 1$, Theorem 1 applies at $N \geq T$ with the preceding $h, \beta = 1/m$; Lemma 2 applies at $C \leq N < T$ with cost J. Taking the maximum of the two budgets gives total expected and high-probability mistake bounds polynomial in $d, C, \log(1/\delta)$, uniformly for every $N \geq C$. These are statements at the specified positive variances. The polynomial's degree may depend on L, γ; this proof makes no uniform claim as $\gamma \downarrow 1$.*

4.4 Mean field

For tanh at $\gamma = 1$, put

$$K = 4A + 8\log(Lm), \qquad T = T_L(d, K), \qquad h = G_L(d, K), \qquad J = AT/2.$$

The same calculation gives relative exceptional mass at most $1/m$ for $N \geq T$. Below T, the direct target cost is at most J.

Corollary 5 (Mean-field cumulative learning)**.** *The finite-variance mean-field tanh learner has total expected and high-probability mistake bounds polynomial in $d, C, \log(1/\delta)$, uniformly over every $N \geq C$, by combining Theorem 1 and Lemma 2 with the preceding budgets. The polynomial degree may depend on depth.*

Why 2^{-d} does not create an exponential sample cost. The relative tail is reduced to 2^{-d} by increasing the structural budget by $O(d)$. The log size of the finite relevant class is polynomial in this budget. One never requests an approximation accuracy of 2^{-d} from a compression theorem whose cost is polynomial in inverse accuracy.

4.5 An extension to ReLU

The ReLU manuscript [6] replaces bounded-activation compression by product-norm capacity control. Its independently reviewed structural bounds can be inserted into the same cumulative theorem. For a concrete depth-L, width-C ReLU margin teacher, set

$$P_* = \prod_{s=1}^{L} \|W_*^s\|_F, \qquad a_H = 1 + \sqrt{2(L-1)\log 2}, \qquad v_* = \lceil 4e(d+1)P_*^2 a_H^2 \rceil.$$

The ReLU strong-prior proof gives a finite relevant class with $h = v_* \log(m+1)$, independent of learner width. Choose the variance small enough at each fixed $N \geq C$ that its relative tail is at most $1/m$. Theorem 1 then gives expected total errors $O(v_* d^2)$, with the stated logarithmic-confidence extension. For a bounded-weight teacher, $P_*^2 \leq (d+1)C^{4L-2}$, so this is polynomial at fixed depth.

There is also a specified finite-variance result for ReLU, without a small-variance limit. The same manuscript proves the following energy tail and count. Define

$$\begin{aligned}
K_R &= L(4dP_*^2)^{1/L}, \qquad S_R = d + L, \\
r_R &= \min\{1, [4\sqrt{2L}(\sqrt{K_R/L} + 1)^{L-1}]^{-1}\}, \\
B_R &= \tfrac{1}{2}(\sqrt{K_R} + r_R)^2 + S_R \log\left(3\max\{1, \sqrt{S_R}/r_R\}\right), \\
T_R &= 4[B_R + (S_R/2)\log 2 + \log m], \\
v_R &= \left\lceil \frac{4(d+1)}{d}(T_R/L)^L a_H^2 \right\rceil.
\end{aligned}$$

For every $N \geq C$ and every $\gamma \geq 2$, the Gaussian with energy coordinate variance $N^{-\gamma}$, globally margin-conditioned as above, has relative function mass at most $1/m$ outside a finite class with $h = v_R \log(m+1)$. Thus Theorem 1 gives an all-width total mistake bound polynomial at fixed depth, uniformly over $\gamma \geq 2$. For one hidden layer, $L = 2$, the same statement holds for every $\gamma \geq 1$. The deeper finite-variance range $1 \leq \gamma < 2$ is not settled by this imported ReLU argument. These assertions use the ReLU paper's homogeneous balancing and capacity proofs, not an extension of the bounded-activation compression theorem by assertion.

5 Depth-polynomial support restrictions

In this section C denotes the width of a layered *Boolean* circuit whose stages fit the learner's hidden layers. Constant wires and any fixed-basis compilation overhead count toward its width and depth. This is the teacher convention of the neuron-merging results; it is a stronger requirement than a comparably small arbitrary real-valued neural teacher.

The same statistical proof applies to finite loss classes, not just Boolean hypotheses. Suppose a posterior of real-output networks admits, apart from posterior mass at most β at every history, a map into a finite collection of losses $\ell_h : \mathcal{X} \to [0,1]$ such that

$$\ell_h(X_i) = 0 \text{ on every training input}, \qquad \mathbf{1}\{f(x)z_w(x) \leq 1/4\} \leq \ell_h(x) \quad \text{for every } x.$$

For a loss with mean u, at least mu inputs have positive loss, so the probability that its empirical loss is zero on a size-n random prefix is at most e^{-nu}. Its remaining mean equals $m/(m-n)$ times its full-domain mean. Every step of Theorem 1 therefore applies unchanged. The map may depend on the retained network and the history; the finite loss class must be fixed before drawing the input order. Here the cumulative posterior error variable uses the tested margin event $fz_w \leq 1/4$; the resulting bounds also control sign errors.

The neuron-merging papers [4, 5] supply exactly this form of deterministic compression. Retained networks have a compressed grid representative whose clipped output differs uniformly by at most $3/32$. Training uses signed margin $> 1/2$, so the representative has signed clipped output $\geq 13/32$ on training inputs. A tested margin error $fz_w \leq 1/4$ gives representative signed clipped output at most $11/32$. Use a ramp loss equal to one at or below $11/32$ and zero at or above $13/32$. The resulting finite class has log size polynomial jointly in $d, L, C, c + c^{-1}, \log(1/\beta)$, independently of N and the scale $0 < t \leq 1$. Choosing $\beta = 1/m$ proves a total mistake bound polynomial in depth as well as width for those structural support restrictions. At fixed remaining parameters, the teacher-width dependence remains quadratic up to logarithms, with an extra factor polynomial in d.

This does not by itself establish a polynomial-depth total mistake bound under the weaker internal-noise restriction alone. Its current entropy compression couples predictions with an error probability α and has size polynomial in $1/\alpha$. Taking α of order 2^{-d} would destroy the desired dimension dependence. That is a limitation of this extension, not a counterexample to such a cumulative theorem. If the weaker support is additionally globally margin-conditioned, the exact finite-function mean-field argument still gives the fixed-depth cumulative result after using its robust target-mass cylinder; it does not preserve the improved polynomial depth dependence.

6 Which input assumptions matter?

Uniform random order is doing real work. A class of $m+1$ simple functions can require m mistakes in a prescribed order despite log cardinality only $\log(m+1)$. Let $f \equiv +1$ and let g_i differ from f only at point x_i. Give g_i prior weight R^{m-i+1} and f weight one, and reveal $x_1, \ldots, x_m$.

Before x_i, g_i has posterior mass greater than $1 - 1/R$, because the previous candidates have been eliminated. For $R > 3$, every round is a 2/3-confidence failure, majority prediction is wrong every time, and expected posterior-sampling mistakes exceed $(1 - 1/R)m$. All functions are individually simple point exceptions on the Boolean cube. This is a counterexample to an arbitrary-order conclusion from a finite relevant-class count alone. It is not a counterexample proved for the particular Gaussian neural-network priors.

An arbitrary iid input distribution is also different from uniform iid. In this same example, take probabilities proportional to $1, \rho, \ldots, \rho^{m-1}$. The chance that the first occurrence of x_{i+1} precedes that of x_i is $\rho/(1 + \rho)$. A union bound shows that the first-appearance order is the displayed adverse order with probability at least $1 - (m - 1)\rho$. Thus even iid sampling can be almost adverse when point probabilities range over sufficiently many scales. Again, this isolates a limitation of the statistical inference rather than proving failure of a specific neural-network prior.

A useful nonuniform extension. For any iid distribution on a finite support of size m and minimum positive probability $p_{\min}$, the same relevant-class assumptions give

$$\mathbb{E}S \le 3 + (h + 1)(1 + \log T) + m\beta, \qquad T = \left\lceil p_{\min}^{-1} \log(2/p_{\min}) \right\rceil .$$

Indeed, expected relevant-class risk after n labels is at most $(h + 1)/n$. It is also at most the expected unseen probability, which is at most $e^{-np_{\min}}$. Sum the first bound through T and the second afterwards. The outside-class contribution is at most β times the unseen probability at every history, and the sum of the expected unseen probabilities equals the support size m. In particular, the total bound stays polynomial whenever $\log(1/p_{\min})$ does. Uniform sampling is a clean special case, not the only useful one.

What this does not show about a converse. Few realized mistakes in one run do not alone imply that the target has a small circuit: favorable random guesses and incomplete domain coverage must be excluded. Nor does a general bounded-mistake learner automatically supply a small circuit unless its predictions have controlled computational descriptions. The results here establish the upper direction for specified Bayesian learners; a converse requires an additional argument.

A Explicit imported neural-network budgets

For reproducibility, the quantities below are the scalar bounds supplied by the earlier proofs. Their mathematical roles are stated in Section 4. They are intentionally loose.

A common norm-sublevel count. For $J \ge 1$, put

$$\begin{aligned} \rho_J &= \frac{2^{-L}(J + 1)^{-(L-1)/2}}{12(L - 1)}, \\ u &= \max\left\{ \lceil 9(J + 1) \rceil, \left\lceil 256\rho_J^{-4}[d + \log(L/\rho_J)] \right\rceil \right\}, \\ F_L(d, J) &= [u(d + 2) + (L - 2)u^2] \log\left(3 + 16Lu \max\{u, d + 1\}[u(u + 1)]^{L-1}\right). \end{aligned}$$

The norm-compression theorem preserves all Boolean signs and yields a width-u, weight-bounded implementation. Uniform parameter rounding gives the count F_L. The class is common to all original learner widths.

The target cylinder. Set

$$\begin{aligned} b_L &= \sqrt{16\log 9 + 8\log(4L)}, \qquad a_0 = 2 + 3C^2 + b_L, \\ \rho_0 &= [2 + 9C^2(d + L + 1)a_0^{L-2}]^{-1}, \\ B_0 &= d(d+1)C^2 + 2(L-2)C^4 + 36C^4, \\ A_L(d, C) &= 2B_0 + 4[d + 2 + (L-2)C]/\rho_0 + 2. \end{aligned}$$

The earlier cylinder proof gives raw target mass at least $\exp(-A_L(d, C)N^\gamma/2)$ for every $\gamma \geq 1, N \geq C$.

Mean-field exact-function compression. Let $r = L - 1$, $K \geq 4$, and choose backwards

$$g_r = 32rK^2, \qquad g_s = 8g_{s+1}(1 + g_{s+1})^2 \quad (s = r-1, \ldots, 1).$$

Put

$$q = \lceil g_1^2/2 \rceil, \quad \ell = \log(2 \cdot 2^{2d} + 1), \quad T_L(d, K) = \lceil \max\{1, 2q\ell/(K-1)\} \rceil .$$

For the finite-description bound, write $g = g_1$, $b = 8(1+g)^2$, and set

$$\begin{aligned} \eta &= [32Krb^{r-1}]^{-1}, & u_* &= \lceil 2\eta^{-2}\log(8r\,2^{2d}) \rceil, \\ D_* &= u_*(d+1) + (L-2)u_*^2, & B_* &= \max\{1, \sqrt{4g + 2\log(8\max\{D_*, 1\})}, 4K/u_*\}, \\ P_* &= D_* + u_*, & \Lambda_* &= (d + Lu_*)[1 + u_*(B_* + 1)]^{L-1}. \end{aligned}$$

Then $G_L(d, K) = P_* \log(3 + 16B_*\Lambda_*)$ is valid. At fixed depth, T_L and G_L have polynomial majorants in d, K. The covariance compression uses the global margin to recover exactly the whole Boolean sign function. It is not merely an average-error approximation.

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